

Power after training and freezing the models

Study III-B | Full-procedure simulation report
HAR versus HAR + current Asian realised variance

Purpose. Replace a shifted-loss planning approximation with a simulation that reconstructs causal HAR features, fits both models, estimates smearing, freezes the fits and generates genuine out-of-sample forecast losses on synthetic sequences.

Planning result

For the calibrated 5% long-training effect target, QLIKE rejection probability is 80.4-83.4% at 2,000 scored synthetic dates and 93.2-95.2% at 4,000, across generator block lengths 5, 10 and 20. The achieved mean effect at 4,000 is 4.64-4.76% after finite 400-row training.

Why this is not a replacement sample-size mandate

The operational-null rejection rate reaches 5.4% at N=8,000 under the block-20 generator (Wilson 95% interval 3.74-7.74%), versus nominal 2.5%. The stationary generator also occasionally violates session/daily RV containment and does not represent all regime shifts or missing labels. These conditional curves cannot certify a uniformly calibrated test or amend the frozen 32,001 requirement.

Simulation element	Executed design
Outer paths	500 per effect/dependence cell; 7,500 total
Training and burn-in	400 fitted rows after 500 synthetic burn-in steps
Scored horizon	250, 500, 1,000, 2,000, 4,000, 8,000
Effect targets	0%, 1%, 3%, 5%, 10%; achieved effects reported
Dependence	Circular blocks 5, 10, 20
Inference	Paired block-bootstrap QLIKE intervals; all blocks

All calibration uses already-viewed 2016-2023 data. No real fresh-period forecasts, trading strategy or 2024+ data are used. This report is separate from the existing Study III Design Pack.

1. The complete simulated procedure

Step	Implementation
1. Generate history	Causal London, Asia and daily variances after a 500-step burn-in
2. Form HAR predictors	Prior London, prior UTC daily, log mean of prior 5 and 20 London RV values
3. Train both models	400 synthetic labelled observations; OLS log RV with intercept
4. Estimate retransformation	Each model uses its own training-only mean $\exp(\text{residual})$ smearing
5. Freeze once	No rolling refit, validation tuning or look-ahead
6. Issue forecasts	HAR and HAR+Asia use past variances and current Asia only
7. Score	Actual synthetic RV and forecast levels produce QLIKE, MSE and MAE
8. Test and repeat	Paired block confidence intervals and secondary gates on each path

Equations and target

HAR regressors $X(t)$ contain $\log L(t-1)$, $\log D(t-1)$, $\log(\text{mean of prior 5 } L)$, and $\log(\text{mean of prior 20 } L)$. The enhanced regression adds $\log A(t)$. Here L , A and D denote positive London, Asian and UTC daily RV. The forecast is $\max[10^{-12}, \exp(\text{fitted log RV}) \times \text{training smearing}]$.

$QLIKE(y, f) = y/f - \log(y/f) - 1$. The paired difference is HAR loss minus HAR+Asia loss: positive values favour Asia. Relative gain is mean paired difference divided by mean HAR QLIKE loss. No synthetic loss is shifted to create the requested effect.

What is counted

A synthetic step is an eligible observation, not a dated market session. Reusing one path at several forecast prefixes provides a power curve; it does not create independent samples. No date outside the allowed real-market interval is attached to a synthetic observation.

2. Dependence-preserving recursive generator

The native-M5 numerical gate passed, so nuisance calibration uses the candidate native-M5 RV method. There are 1,904 usable pilot rows from 2016-02-01 to 2023-12-29 with HAR inputs and current Asia, London and daily RV. Pilot feature eligibility uses only information available before London; the additional current daily value is used solely to fit the synthetic nuisance generator after the fact.

Regress log Asia on HAR inputs and obtain residual $u(t)$. Regress log London on HAR plus log Asia. Re-express London as a reduced HAR component plus the fitted Asia coefficient times $u(t)$, plus London residual $e(t)$. The partial Asia coefficient in this nuisance fit is 0.32947; it is calibration, not a newly validated empirical result.

Bootstrap $u(t)$ in one circular block stream. Independently bootstrap the joint vector $[e(t), \log(D(t)/L(t))]$ in another. Generate Asia from its HAR equation and $u(t)$; generate London from reduced HAR, signal multiplier times the Asia innovation, and $e(t)$; generate $D(t)$ by multiplying $L(t)$ by the resampled positive ratio. Every subsequent HAR input is recomputed from these generated past values.

Generator property	Observed / imposed
Pilot residual lag-1 correlation: Asia	0.0849
Pilot residual lag-1 correlation: London	0.0144
Pilot daily/London log-ratio lag-1 correlation	0.0736
Reduced-HAR absolute lag-coefficient sum	0.94321; below preregistered 0.97 cap
Asia absolute lag-coefficient sum	0.91863; below preregistered 1.5 cap
Coefficient shrinkage applied	None; intercept reset differs only by numerical precision
Pilot daily RV below London RV	0 rows

Blocks preserve within-stream serial dependence and London-residual/daily-ratio dependence. Independent streams intentionally remove their cross-stream association to control the Asia channel. This is not preservation of the entire empirical joint law. At target 5% and block 10, synthetic log-RV lag-1 correlations are 0.749 London and 0.677 Asia, versus 0.624 and 0.622 on eligible pilot dates.

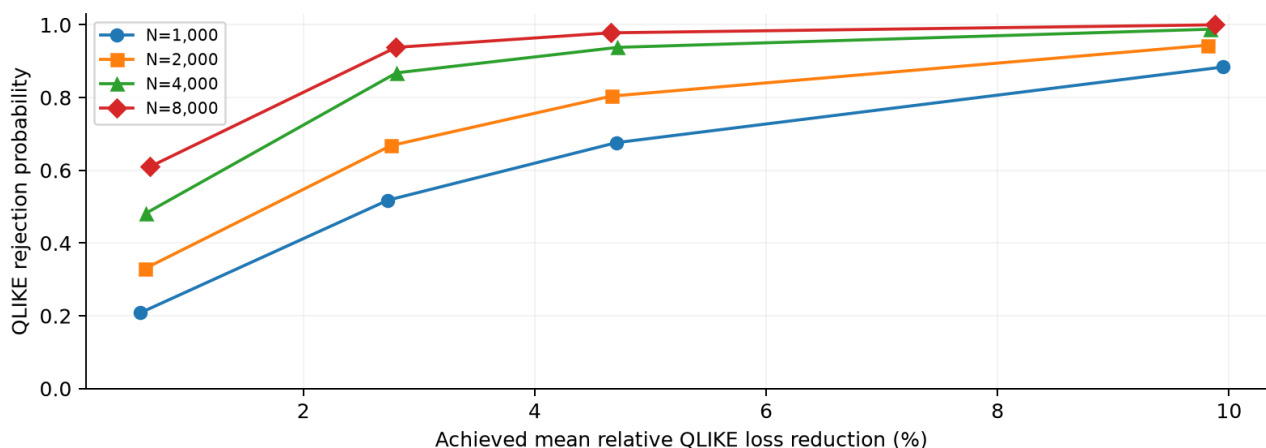
Post-run realism diagnostic. In separately seeded 128-path checks per cell, Asia exceeds daily RV on 0.37-0.38% of steps at the 5% target (0.74-0.75% with no direct signal). The generator is a reduced-form variance model, not a coherent intraday price path. No offending path was filtered and no primary result was replaced. A coherent-component sensitivity is needed before adopting a new N.

3. Effect calibration and training uncertainty

With an independent random seed, nine signal multipliers from 0 to 3 were evaluated on 16 synthetic paths each, with 4,000 training rows and 8,000 scored observations under block-10 dependence. The first upward crossing of each target gain was linearly interpolated. All four targets were bracketed without expanding the frozen grid.

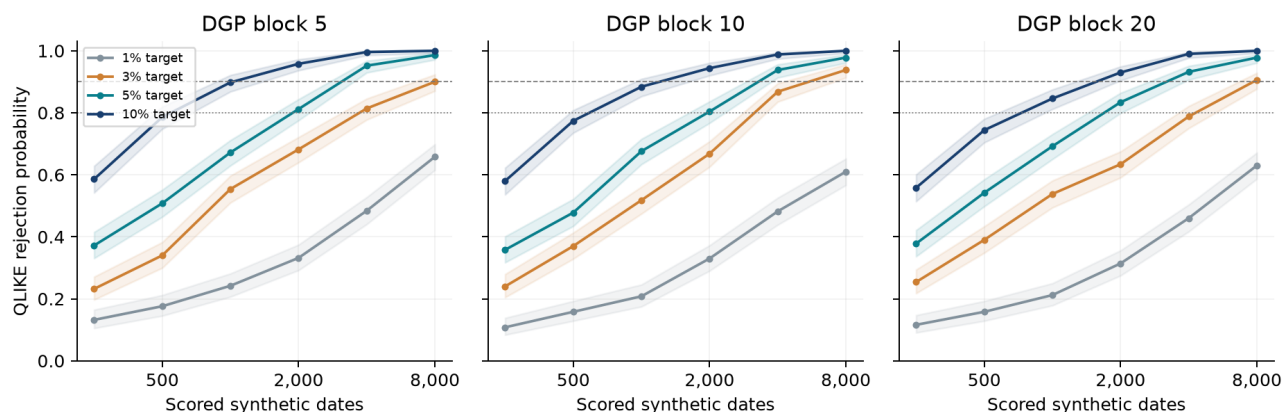
Target gain	Mean achieved gain, N=4,000	Rejection probability
0.0%	-0.2%	2.0%
1.0%	0.6%	48.2%
3.0%	2.8%	86.8%
5.0%	4.7%	93.8%
10.0%	9.8%	98.8%

Table: block-10 generator; 500 independent 400-training-row paths. Calibration targets describe the long-training experiment. The finite-training achieved gain is the relevant reported effect, and can be smaller or negative on individual paths.



At the 5% target, block-10 and N=4,000, achieved effects span 2.51-7.21% between the 5th and 95th path percentiles. The mean is 4.717%, with Monte Carlo standard error 0.066 percentage points. Uncertainty from calibration-parameter estimation is not included in that Monte Carlo standard error.

4. Power curves across effect sizes



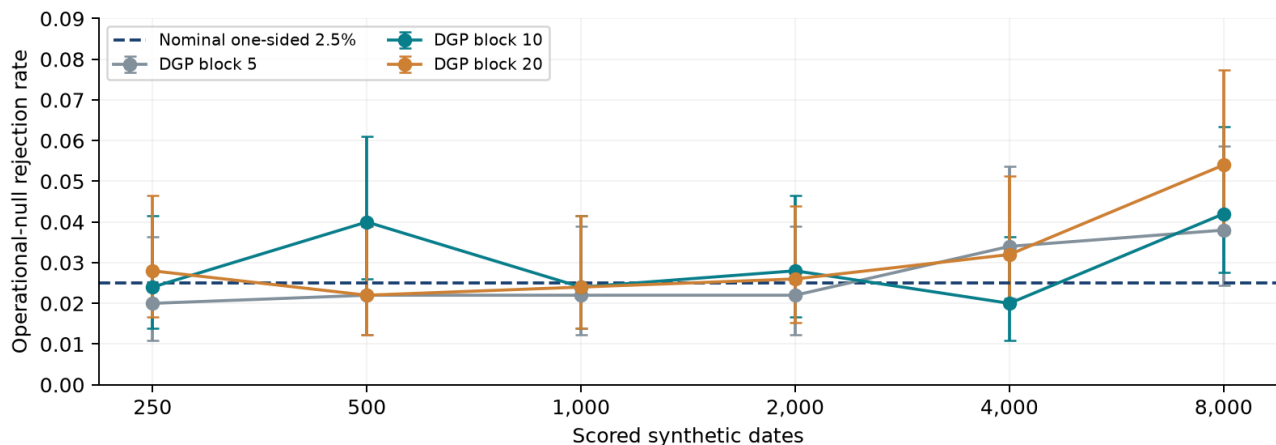
Shaded bands are Wilson 95% Monte Carlo intervals conditional on the fitted generator and the shared independent inner-bootstrap resampling matrices. They do not include uncertainty about the real market, the nuisance DGP or future feed quality.

DGP block	Scored N	5% target power	95% MC interval	Achieved gain
5	2,000	81.2%	77.5% - 84.4%	4.7%
10	2,000	80.4%	76.7% - 83.6%	4.7%
20	2,000	83.4%	79.9% - 86.4%	4.8%
5	4,000	95.2%	93.0% - 96.8%	4.8%
10	4,000	93.8%	91.3% - 95.6%	4.7%
20	4,000	93.2%	90.6% - 95.1%	4.6%

The 5% target first crosses 80% point-estimated power at 2,000 and 90% at 4,000 in all three dependence settings. At 2,000, none of the lower Monte Carlo bounds reaches 80%; at 4,000 all lower bounds exceed 90%. These are grid crossings, not interpolated exact minimum sample sizes.

For 1%, neither 80% nor 90% is reached by 8,000. For 3%, 90% point-estimated power first appears at 8,000 in all settings, but the lower bounds do not all support 90%. For 10%, 90% point-estimated power is first reached at 2,000. Complete grid crossings, including lower-bound crossings, are supplied in GRID_CROSSINGS.csv.

5. Operational null and inference checks



Zero multiplier removes the direct incremental Asia channel. It does not impose exact equality of the two finite-trained forecast losses: estimating an extra regressor can help or hurt a particular frozen pair. Residual block dependence and model approximation also remain. Therefore this is an operational no-channel null diagnostic, not an exact least-favourable equal-loss size proof.

At $N=4,000$, null rejection is 3.4%, 2.0% and 3.2% for DGP blocks 5, 10 and 20. At $N=8,000$ it is 3.8%, 4.2% and 5.4%. The latter two Wilson intervals exclude 2.5%. The procedure should not be advertised as uniformly nominally calibrated on this evidence.

Bootstrap resolution check

The primary planning surface uses 1,999 circular paired-block bootstrap draws and requires a positive lower 2.5th percentile for every inference block 5/10/20. At target 5%, DGP block 10 and $N=4,000$, 10,000 draws give 94.2% power versus 93.8% on the planning surface: a 0.4 percentage-point difference. No primary curve was overwritten.

HAC-20 is recorded as a descriptive sensitivity diagnostic. Clark-West adjustment is not applied mechanically to QLIKE; its nested linear squared-error setting is different. The actual test evaluates the frozen trained forecast pair conditional on training, while the outer simulation integrates variation across training samples. [2, 3]

6. Statistical power is not study acceptance

A secondary simulation statistic additionally requires positive fitted Asia coefficient and no more than 5% MSE or MAE deterioration. At the 5% target and N=4,000, joint rejection equals QLIKE rejection in all three dependence settings: 95.2%, 93.8% and 93.2%. This does not imply that every real-study acceptance gate would pass.

Captured in this simulation	Not established by this simulation
400-row estimation and frozen forecasts	True out-of-period gold confirmation
Training-only smearing and nonlinear losses	Native M5 proxy unbiasedness on sparse quotes
Serially dependent nuisance blocks	All future regime shifts and microstructure changes
Paired block confidence intervals	Independent-provider agreement
Coefficient and secondary-loss gates	95% real issued-forecast label coverage
Finite training uncertainty	Each full real-calendar-year consistency gate

Relationship to the original power calculation

The old 32,001 figure came from a conservative paired-loss long-run-variance envelope and a shifted-loss simulation. The new curves answer a different, more mechanistic conditional question: what happens under this calibrated recursive process after estimating and freezing both models? Differences in N reflect changed modelling assumptions, not a discovery that the old freeze can be ignored.

Design implication

A 4,000-scored-date design is a useful candidate for further planning under the present 5% calibration, not an approved replacement requirement. Null behaviour, a coherent session/daily variance generator, calibration uncertainty, real calendar-year gates and actual fresh-history coverage must be resolved in a new registered design before selecting an acceptance rule or evaluating fresh outcomes.

No data acquisition before 2016 was performed. Better native-M5 coverage in already-viewed dates does not create fresh outcomes, and repeated broker observations of the same dates do not multiply independent market information. The 2024+ holdout remains unopened.

7. Reproduction, validation and references

The full simulation algorithm was frozen before execution in FULL_PROCEDURE_SIMULATION.md, commit 875b4e0. The executable model and causal-timing tests were committed before simulation in 3ab9eb5. All market inputs are 2016-2023. Seeds are 314159 for effect calibration and 271828 plus fixed cell offsets for the main synthetic paths.

Artifact	Contents
DGP_CALIBRATION.json	Nuisance fits, stability checks and pilot properties
PILOT_FEATURE_LEDGER.csv	Causal feature and label eligibility
EFFECT_CALIBRATION_GRID.csv	Signal multipliers and achieved long-training gains
POWER_CURVES.csv	90 cells: effects x dependence x scored N
MONTE_CARLO_REPLICATE_LEDGER.csv	45,000 path-prefix decisions and interval endpoints
GRID_CROSSINGS.csv	Point and Monte Carlo lower-bound 80/90% crossings
INNER_BOOTSTRAP_CONVERGENCE.json	10,000-draw sensitivity at the frozen check cell
SIMULATION_RUN.json	Run dimensions, elapsed time and scope

Tests verify the date guard, London DST, omission of returns across gaps, causal recursive HAR lags, unchanged fitted forecasts when future outcomes change, and equivalence of count-matrix circular bootstrapping to explicit sampled indices. All tests pass. The original M1 RV is reproduced exactly on frozen valid sessions in the companion audit.

References

- [1] Patton, A. J. (2011). Volatility forecast comparison using imperfect volatility proxies. Journal of Econometrics 160, 246-256. [Author-hosted paper](#).
- [2] Newey, W. K. and West, K. D. (1987). A simple, positive semi-definite, heteroskedasticity and autocorrelation consistent covariance matrix. [NBER working paper](#).
- [3] Clark, T. E. and West, K. D. (2007). Approximately normal tests for equal predictive accuracy in nested models. Journal of Econometrics 138, 291-311. [Federal Reserve working-paper version](#).
- [4] Study III-B frozen simulation method and native-M5 equivalence report; original Study III Design Pack retained without amendment.